

L09 OCCUPANT LIGHTING CONTROL | O (MAX : 3 PT)

Intent : Provide individuals with access to customizable lighting environments.

Summary : This WELL feature requires projects to implement innovative lighting strategies that take into account personal preferences of users, as well as their interaction with the physical space.

Issue : Humans spend about 90% of their time indoors and the ambience of the indoor environments has an impact on the well-being and productivity of occupants. Lighting in a space has been shown to have a positive impact on mood and cognitive performance.^{1,2} Lighting environments that are customizable by individuals have been shown to improve satisfaction levels.³ The ages of occupants have an impact on the light levels required for visual acuity.^{4,5} Lighting guidelines recommend twice the light levels for individuals over the age of 65 years, compared to individuals between the ages of 25 years and 65 years.^{5,6} As humans age, the transmission of light through their lens is reduced. This is due to age related changes, including increased light absorption by the lenses smaller pupil size, increased scattering of light due to thicker lenses and yellowing of the lenses.^{7,8}

Solutions : Developing a lighting environment that not only seeks to satisfy the visual and circadian requirements of individuals, but also creates a customizable environment helps to improve productivity, mood and well-being. Innovative lighting strategies, including the customization of a user's immediate environment, can contribute to occupant satisfaction with the space. Creating zones with lighting conditions that are distinct from the lighting in regular workspaces, can create a comfortable and informal environment that individuals can utilize for social interaction. Lighting environments can help to improve mental health, reduce stress and improve visual acuity.^{1,9}

PART 1 ENHANCE OCCUPANT CONTROLLABILITY (MAX : 2 PT)

For All Spaces:

1: Lighting zones

Ambient lighting systems meet the following requirement:

a.

All regularly occupied spaces contain lighting zones as shown in the table below (note: individual rooms smaller than the areas below and/or that have occupancies less than those listed in the table are considered separate zones):

Tier	Number of Zones		Number of Zones	Points
1	One per 60 m ²	OR	One per 10 occupants	1
2	One per 30 m ²	OR	One per 5 occupants	2

2: Lighting control system

Each lighting zone meets the following requirements:

- Lighting systems have at least three lighting levels or scenes that allow for changes in light levels.
- Lighting systems have the ability to change at least one of the following:
 - Color.
 - Correlated color temperature (CCT).
 - Distribution of light by controlling different groups of lights or through preset scenes.
- All regular occupants have control over their immediate lighting environment through at least one of the following:
 - Manual controls (e.g. switches or control panels) located in the same space as each lighting zone.
 - Digital interface available on a computer or phone.
- Lighting for presentation or projection walls are separately controlled.

Note :

Projects may identify spaces within the project boundary where lighting controls are limited to certain regular occupants (e.g., facilities staff, office manager) provided that these spaces do not make up more than 10% of the project size.

PART 2 PROVIDE SUPPLEMENTAL LIGHTING (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Supplemental lighting availability

The project meets the following requirements:

- Supplemental light fixtures (e.g., task lights) are provided upon request to all employees at no cost and requests are fulfilled within eight weeks.
- At least one supplemental light fixture is available for trial purposes.

2: Supplemental lighting requirements

Supplemental light fixtures meet the following requirements:

- Light levels are controllable by occupants, independently from the ambient lighting system.

- b. The location of the light is adjustable by users of the workstation.
- c. Under intended use, the light-emitting element is not visible to users.

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